

Time-Frequency Analysis of EEG Signals from Healthy Subjects Allocated by Gender for a Subject-Independent BCI-Based on Motor Imagery

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Abstract— Most of the recent brain-computer interfaces (BCI) based on motor imagery are designed by taking into account a single user. A BCI system designed for multiple users without the need of extensive training sessions could be a viable solution for the system's implementation outside research centers. The present work explores the design of an independent-subject BCI. In order to accomplish this, Linear Discriminant Analysis classifiers were designed with the data of a sample of 30 healthy volunteers as a whole group and separated by gender. Three different methods were employed to compute power spectrum features from the volunteer's electroencephalographic recordings. The results show that it is possible to design an independent-subject BCI for the classification of right or left hand motor imagery with respect of a reference interval with classification accuracies above 70%. The female gender could benefit more from a subject-independent classifier, than the male gender.

I. INTRODUCTION

Brain-computer interface (BCI) based on motor imagery are devices that translate electrical brain signals, commonly recorded through electroencephalography (EEG), in commands for controlling orthoses, prostheses or robotic devices [1]. BCIs have potential as a tool for aiding patients with neuromotor diseases with their rehabilitation. Nevertheless, their use outside of research centers is very limited, because a significant quantity of recordings per subject is necessary to generate a classifier and obtain a BCI with good performance for that single user [2]. Most of the patients that could be benefitted from a BCI do not have the availability, either because of their physical or psychological conditions, to fulfill the required stages needed for the system implementation. Therefore, an independent-subject BCI (trained with the data of a subject group instead of only one subject) that allows a new user to obtain good performance with minimal training could increase the benefits for patients with neuromotor disabilities.

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This work presents the EEG signal analysis of a sample of 30 healthy subjects while performing motor imagination of the opening and closing of the right and left hand, with the objective of determining if in a design of an independent-subject BCI, the classification accuracy increases by training a classifier with data from a sample of the same gender. Three different types of time-frequency analysis are used to compute spectral features and different groups of electrodes are used to evaluate the performance of a linear discriminant analysis (LDA) classifier.

II. METHODS

Here we describe the protocol used to record the EEG signals, the feature extraction and the implementation of a classifier used as base for the design of an independent-subject BCI. Feature extraction was defined using previous knowledge, which showed that the power spectrum normalized to a reference interval, decreases in the range of alpha (8-13 Hz) and beta frequencies (14-30 Hz), when the subject imagines movement of his extremities. These changes in power are recorded, mainly, in the central channels placed above the motor and sensory cortex [3] [4] [5]. The observed decrease in power spectrum when performing such tasks is known as event related desynchronization (ERD) [6].

A. Subjects

The sample consists of 30 healthy participants in ages ranging from 21 to 30 years, 50% from the female gender and 50% from the male gender (Mean=25.8 years and SD=2.94 years), with incomplete or completed bachelor degrees. All are right handed, with normal or corrected to normal vision and without antecedents of neurological lesions. An expert in electrophysiology evaluated the EEG recordings to confirm the subject's neurological condition. Before the study began, all subjects signed an informed consent approved by the ethics and investigation committee of the National Institute of Rehabilitation. All participants achieved a normal performance in the subscales of detection of digits and visual detection, as well as the neuropsychological instrument NEUROPSI Attention and Memory which evaluates the capacity to follow instructions and attain repetitive tasks without being distracted by other stimuli [7]. The subjects had no previous knowledge of the experiment, or training to perform the imagery tasks, which were explained with verbal instructions minutes before initiating the recording.

B. Experimental task

The subject was sat in a comfortable chair with arms in rest, and a computer screen of 15.4" was placed in front of him at 150 cm. The EEG recording consisted of a series of trials with a duration of 8 s that initiate with the presentation of a white fixation cross in the center of a black background, followed by a short duration warning tone at 2 s. At the third second, the cross was overlapped by an arrow pointing at either the left or the right side of the screen during 1.5 s. Depending on the arrow's direction the subject imagined the continuous movement of opening and closing the right or left hand, until a blue background appeared without any visual guide, during a random period between 3 and 5 s, in which the subject was instructed to rest and blink his eyes. The trial was repeated 20 times, the right and left arrow appeared randomly ten times each, in order to avoid subject habituation. The experiment was based on the Graz paradigm [5].

C. Signal acquisition

EEG signals were recorded with a Nicolet® amplifier model NicONE with 32 channels and 16-bits resolution, with a sample rate of 256 Hz. A total number of 20 electrodes where placed in the subject's scalp following the international 10-20 system. The ground and reference electrodes were placed in the central forehead line. Electrooculogram (EOG) was also measured using two electrodes placed on the orbiculus oculi muscle on both eyes to record eye movements, and in the masseter muscles to detect signal artifacts. The electrode impedances were kept below 5 K Ω . Electromyography (EMG) electrodes were placed in the subject's arms (above the deep flexor and superficial muscles of the fingers) to ensure the absence of movement in hand motor imagery tasks.

D. Signal Pre-processing

The EEG signals were recorded in referential configuration and were conditioned with a 40 Hz low-pass filter, a 1.6 Hz high-pass filter and a band-stop filter of 59-61 Hz. An independent components analysis (ICA) was used to eliminate eye, muscular and hearth activity artifacts. Independent components were calculated based on 20 EEG, 2 EOG and 2 EMG channels. Components containing artifacts were set to zero and the signal was converted back into EEG domain. The detailed procedure can be found in the work of Jung, et al. [8]. All the recordings were read, pre-processed, and processed using the software Matlab® R2012b from Mathworks® and the free license toolbox Fieldtrip [9].

E. Time-frequency analysis

Spectral features of the EEG signals from all the subjects were obtained by using three time-frequency analysis: the Fast Fourier Transform (FFT) using a fixed length time window of 300 ms, the FFT using a variable length time window which decreases with frequency (both FFT were implemented with time windows of Hanning type displaced

and overlapped every 50 ms) [10], and the Morlet type Wavelet transform (WT). All the analysis were made from 1 to 7 s and from 1 to 30 Hz with a resolution of 0.5 Hz. By using every technique (WT, FFT with variable and fixed window, and with all electrode combinations) a time frequency representation (TFR) was computed for every subject, and normalized to a reference interval which ranged from 1.5 to 2.5 s.

F. Feature extraction

From the TFRs of every subject, windows of one second of length from 3.5 s to 4.5 s were extracted for every motor imagery tasks of right hand and left hand and from 1.5 s to 2.5 s for its respective reference interval in the 20 recorded channels, with the spectral power in this time windows being averaged. This power spectrum values where again averaged in the 8 to 10.5 Hz and in the 11 to 13.5 Hz range, therefore two features where obtained for each channel.

For the classification, four arrays of five electrodes were selected: C3, Cz, C4, T3, T4; C3, Cz, C4, Fz, Pz; P3, Pz, P4, T5, T6 y F3, Fz, F4, F7, F8. Central and frontal electrodes were favored when selecting the groups because they were located over the motor and sensory cortex.

G. LDA classification

A LDA classifier was selected since it has shown similar performance to other non-linear methods and has the advantage in terms of a lower computational cost [11]. Therefore LDA was considered a good choice for the design of a subject-independent BCI, which was implemented by following the scheme of using the features of several subjects as training data, simulating data obtained from a single subject as it is traditional with subject-dependent BCIs.

A classifier was implemented and tested with the data of the 30 subjects; later that data was separated by the subject's gender and two new classifiers where designed and tested. The classification was made for the right hand motor imagery (RIGHT-MI) vs. reference (REF), left hand motor imagery (LEFT-MI) vs. REF and RIGHT-MI vs. LEFT-MI.

H. Cross Validation

To validate the implementation of the classifiers, the "leave-one-subject-out" method was used, meaning that mixed data from all except one subject was used for the training of the classifier, while the data from the remaining subject was used to test the classifier. In other words, the classifier was tested with one subject whose classifying training data did not include its own recordings, but consisted of the recorded data from the remaining subjects. The training data pool was randomly mixed ten times and each training pool was used to set up a classifier that was tested with the excluded data from one subject, the classification accuracy of the ten repetitions was then averaged. Later, the

procedure was repeated leaving out each of the subjects and the classifying percentages for each test run were averaged.

I. Statistical analysis

The statistic paired-sample Student-t test with a level of confidence of 95% was used to determine if there was a statistically significant difference between the classification accuracies obtained by the designed classifier with the data from the 30 subjects, and the classification accuracy that was achieved with the classifiers designed by taking gender into account. The test was also applied to determine if there was a statistically significant difference between the classification accuracies of the designed classifiers with the different methods for computing spectral power.

III. RESULTS

Fig. 1 shows topographic representations of the grand average of the TFRs for every time-frequency analysis for all subjects. We can observe that in right hand motor imagery a bilateral ERD is displayed; while for left hand motor imagery the ERD is contralateral.

Table 1 and Table 2 show the mean and standard deviation of the percentages of classification accuracy for the four arrays of electrodes that were evaluated when the conditions of LEFT-MI vs. REF and RIGHT-MI vs. REF were made in each case. Each of the tables show the values for the three groups of subjects.

The comparisons of LEFT-MI vs. RIGHT-MI are not shown in the results because for all features in all three conditions, the classification accuracy was below that of a random decision [12], since they are in the range of 45 to 62%.

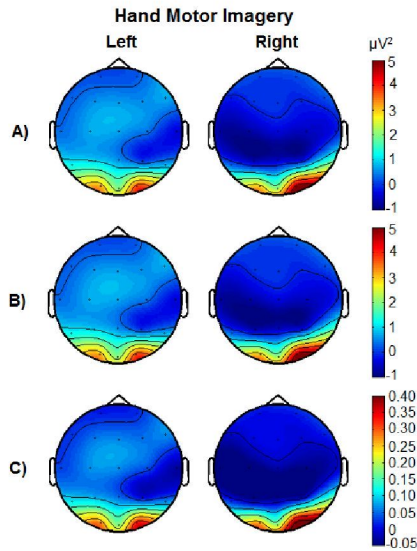


Figure 1. Topographic representation of the grand average of power changes for all subjects in the mu rhythm band. The values (3-7 s) are normalized with respect to a baseline (1.5-2.5 s). A) FFT with a fixed length time window. B) FFT with a variable length time window. C) Wavelet transform.

TABLE I. CLASSIFICATION ACCURACY (%) BETWEEN LEFT HAND MOTOR IMAGERY AND REFERENCE INTERVAL. THE BEST ACCURACY FOR GENDER IS DISPLAYED IN BOLD.

Electrodes		C3 Cz C4 T3 T4		C3 Cz C4 Fz Pz		P3 Pz P4 T5 T6		F3 Fz F4 F7 F8	
Method	Group	Mean	SD	Mean	SD	Mean	SD	Mean	SD
FFT-F _a	Male	70.33	10.4	71	11.9	65	9.5	71	9.5
	All-M	70.17	12.4	71.67	13.6	69.67	11.2	72	10.7
	Female	74.67	10.2	70.67	10.8	72	11.2	73.67	9.9
FFT-V _b	Male	67.33	6.4	68.17	9.3	57.50	10.3	75.83	7.5
	All-M	69.67	12.0	71.33	13.2	69.83	11.5	71.83	10.7
	Female	74.17	8.0	75.33	8.9	73.5	8.0	74.33	6.5
WT _c	Male	70.83	6.8	62.67	11.4	61.5	9.2	73.17	6.6
	All-M	67.67	12.7	70.17	13.9	66.67	10.8	70.5	9.5
	Female	73	9.7	77.67	11.0	73.5	8.5	76.17	8.2

a. Fast Fourier Transform with a fixed length time window. b. Fast Fourier Transform with a variable length time window. c. Wavelet Transform

In LEFT-MI vs. REF for all subject groups no significant differences ($p > 0.05$) were found between the classification accuracy for each feature extraction method. In the case of RIGHT-MI vs. REF significant differences ($p < 0.05$) were only found for the female gender between the WT and FFT-variable window methods with FFT-fixed window.

In the test that aims to find classification differences between genders (which probability values are shown in Table 3) it was found that for LEFT-MI vs. REF significant differences ($p < 0.05$) exist between the female gender and the complete group, for the 3 techniques. RIGHT-MI vs. REF presents a statistically significant difference ($p < 0.05$) between male and females groups, the same with women and the complete group in the three techniques used to obtain the power spectrum.

IV. CONCLUSION

For a subject-independent classifier, the power spectrum computing methods that provide better results, classification accuracies above 70%, are the ones that used a variable length time window (used in methods like Wavelet transform, and FFT).

TABLE II. CLASSIFICATION ACCURACY (%) BETWEEN RIGHT HAND MOTOR IMAGERY AND REFERENCE INTERVAL. THE BEST ACCURACY FOR GENDER IS DISPLAYED IN BOLD.

Electrodes		C3 Cz C4 T3 T4		C3 Cz C4 Fz Pz		P3 Pz P4 T5 T6		F3 Fz F4 F7 F8	
Method	Group	Mean	SD	Mean	SD	Mean	SD	Mean	SD
FFT-F _a	Male	72.33	12.2	70.33	9.2	70.67	11.2	73	10
	All	74.33	13.3	73.17	14.8	71	13.7	77	11.3
	Female	76.33	11.8	75.33	11.5	74.33	8.9	77.33	7.0
FFT-V _b	Male	71.5	8.4	69.83	6.9	73.17	8.5	68.33	7.4
	All	74.17	13.6	77.17	14.7	71	14.0	73.83	12.6
	Female	88.00	14.4	85.17	12.7	82.17	10.1	81.5	5.3
WT _c	Male	67.5	10.3	64.33	8.0	72.5	8.0	69.33	9.8
	All	73.5	15.5	71.83	15.6	69.67	12.5	75.67	12.3
	Female	87.67	14.7	84.83	12.6	81.83	10.5	80.33	8.1

a. Fast Fourier Transform with a fixed length time window. b. Fast Fourier Transform with a variable length time window. c. Wavelet Transform

TABLE III. PROBABILITY COMPUTED WITH T-TEST TO DETERMINE STATISTICAL SIGNIFICANCE BETWEEN THE CLASSIFIERS TRAINED WITH THE FEATURES COMPUTED FOR THE GROUPS OF SUBJECTS. THE STATISTICALLY SIGNIFICANT DIFFERENCES ($p < 0.05$) ARE DISPLAYED IN BOLD.

Groups	LEFT-MI vs REF			RIGHT-MI vs REF		
	<i>FFT-F</i> ^a	<i>FFT-V</i> ^b	<i>TW</i> ^c	<i>FFT-F</i> ^a	<i>FFT-V</i> ^b	<i>TW</i> ^c
Male/Female	0.1123	0.1402	0.1173	6.6x10⁻⁵	0.0038	0.0141
Male/All	0.2444	0.3795	0.5968	0.0591	0.2057	0.1723
Female/All	0.0465	0.0033	0.0042	0.0500	0.0060	0.0145

a. Fast Fourier Transform with a fixed length time window. b. Fast Fourier Transform with a variable length time window. c. Wavelet Transform

A subject-independent classifier, trained with a relatively small sample, can differentiate correctly between conditions of motor imagery and a reference period but is unable to do it in conditions of RIGHT-MI vs. LEFT-MI. It could be possible that with the data of a larger sample used for the training of the classifier, or by using other feature extraction and classification methods, classification accuracy of RIGHT-MI vs. LEFT-MI can be improved.

The fact that the female group presented better classification accuracy (from 70% to 88%) in both motor imagery tasks is evidence that common power spectrum features exist, in the alpha frequency band, for the planning and imagination of the motor tasks between the subjects of this gender. On the other hand, classification accuracies were not as high (from 57% to 75%) for the male gender in both motor imagery tasks, perhaps due to the common planning and imagination power spectrum features being coded in other frequency bands rather than in the alpha band. Hence, the female gender could benefit more from a subject-independent classifier, trained with few samples, than the male gender.

Electrodes located in the frontal area showed the best classification accuracies for most of the male sample tests, this could be due to, for the male gender, motor imagery tasks activate more the pre-motor cortex than the motor cortex.

The averaged classification accuracies, for RIGHT-MI vs. LEFT-MI tasks, are superior to 70% which is good enough to control a BCI [13], therefore this work suggests the possibility of designing subject-independent BCI systems.

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